

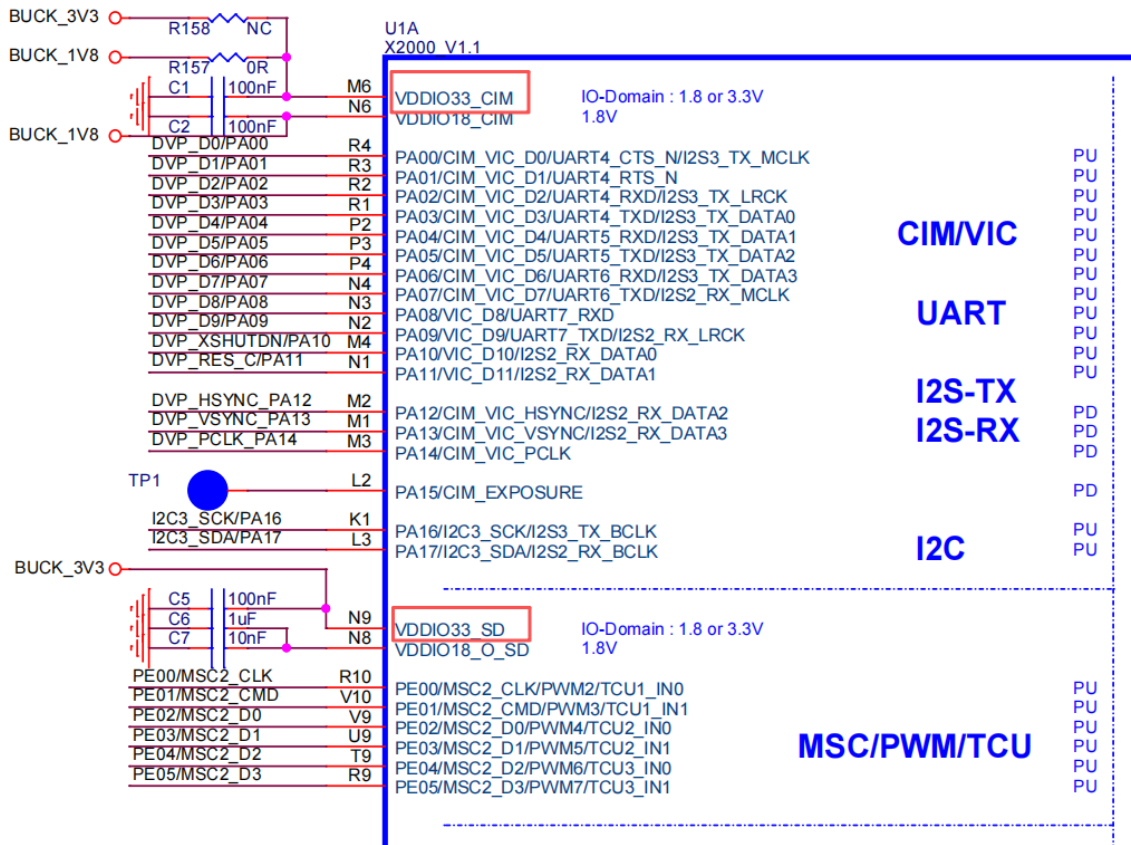
x2000_darwin_rtos_camera_双摄

一 硬件环境

Darwin_X2000_V2.0 : X2000H + nand,uart2_pd31

默认双摄sensor: 两个GC2053, mipi接口。

根据硬件设计可知, 当前VDDIO33_CIM接1.8V, VDDIO33_SD接3.3v。在软件上也要对应设置, 否则长时间使用会有烧坏主控芯片的风险。



二 配置sensor相关

本文档使用x2000的最小编译配置x2000_nand_defconfig, 展示如何配置和使用摄像头。通过isp获取sensor0的帧, 并打印获取的帧数。同时将sensor1的数据通过isp预览到屏幕上。



因双摄需要申请比较大的内存, 所以此处需扩大rtos的固件使用内存大小:

< [root] - [通用选项]

spl bin文件=xburst2/soc-x2000/spl/x2000-spl-sfc-nand-1200M-750M.bin	==>
☐ 生成压缩格式的 rtos	###
工具链路径=../tools/toolchains/mips-xburst2-newlib430/bin	==>
gcc 工具链的前缀=mips-sde-elf-	==>
固件加载的地址=0x86000000	==>
固件使用的内存大小=32777216	==>
异常向量表大小=1024	==>
初始化栈大小=16384	==>
异常栈大小=2048	==>

配置sensor外设：

IConfigTool

File Edit

< [root] - [外设] - [摄像头设备列表]

☐ sensor ov9281 (mipi, raw10)	###
☐ sensor ov9284 (dvp, raw8)	###
☐ sensor gc1054 (dvp raw10)	###
☒ sensor gc2053 (mipi, raw10)	==>
☐ sensor gc2053 (dvp raw10)	###
☐ sensor jz0378_dvp (dvp, grey)	###

< [root] - [外设] - [摄像头设备列表] - [sensor gc2053 (mipi, raw10)]

☒ sensor0 config	==>
☒ sensor1 config	==>

<

[root] - [外设] - [摄像头设备列表] - [sensor gc2053 (mipi, raw10)] - [sensor0 config]

选择索引号(sensor 连接soc控制器接口)=0	==>
gpio power(sensor 模块电源控制引脚)=-1	==>
regulator name(通过名字获取 sensor module 电源管理器)="regulator_gpio0"	==>
gpio pwn(sensor 电源控制引脚)=GPIO_PB(18)	==>
gpio reset(sensor 复位脚)=GPIO_PB(12)	==>
i2c bus num (sensor 挂接的i2c总线号)=5	==>
☒ sensor isp bin select	
sensor isp bin path="devices/camera/x2000/gc2053/gc2053-ir-x2000.bin"	==>

<

[root] - [外设] - [摄像头设备列表] - [sensor gc2053 (mipi, raw10)] - [sensor1 config]

选择索引号(sensor 连接soc控制器接口)=1	==>
gpio power(sensor 模块电源控制引脚)=-1	==>
regulator name(通过名字获取 sensor module 电源管理器)="regulator_gpio0"	==>
gpio pwn(sensor 电源控制引脚)=GPIO_PB(19)	==>
gpio reset(sensor 复位脚)=GPIO_PB(20)	==>
i2c bus num (sensor 挂接的i2c总线号)=2	==>
☒ sensor isp bin select	
sensor isp bin path="devices/camera/x2000/gc2053/gc2053-x2000.bin"	==>

配置sensor对应的电源控制引脚:

<

[root] - [外设] - [设备电源管理] - [gpios regulator(gpio控制电源)]

☒ gpio_regulator0	
regulator0 引脚及有效电平=GPIO_PC(21),1	==>
regulator0 名字="regulator_gpio0"	==>
☐ gpio_regulator1	
☐ gpio_regulator2	

配置sensor对应的i2c:

< [root] - [xburst2系列cpu] - [x2000系列 cpu 设置] - [I2C 总线]

☐ 选择 I2C1

☒ 选择 I2C2

选择I2C2 GPIO=CONFIG_X2000_I2C2_PB ==>

I2C2 速率=400000 ==>

☐ 选择 I2C3

☐ 选择 I2C4

☒ 选择 I2C5

选择I2C5 GPIO=CONFIG_X2000_I2C5_PD ==>

I2C5 速率=400000 ==>

配置vic控制器相关：

< [root] - [xburst2系列cpu] - [x2000系列 cpu 设置] - [camera 驱动 (vic/cim控制器)]

☒ 选择VIC0控制器

选择VIC0 数据通路(ISP/DDR)=CONFIG_SOC_X2000_CAMERA_VIC0_ROUTINE_ISP ==>

VIC0 MCLK管脚选择=GPIO_PE(24) ==>

VIC0 帧缓冲数量=2 ==>

☒ 选择VIC1控制器

选择VIC1 数据通路(ISP/DDR)=CONFIG_SOC_X2000_CAMERA_VIC1_ROUTINE_ISP ==>

VIC1 MCLK管脚选择=GPIO_PE(24) ==>

VIC1 帧缓冲数量=2 ==>

☐ 选择CIM控制器

三 配置屏幕显示相关

< [root] - [外设] - [lcd设备列表] - [TFT FW050 (720*1280) (X2000) (mipi接口)]

gpio backlight enable(lcd backlight使能引脚)=-1,0 ==>

gpio reset level (lcd 复位引脚及有效电平)=GPIO_PB(8),0 ==>

gpio power level (lcd power使能引脚及有效电平)=GPIO_PB(9),0 ==>

regulator name(通过名字获取 LCD module 电源管理器)="-1" ==>

< [root] - [外设] - [背光设备列表] - [GPIO背光]

☒ gpio_backlight0	
GPIO 引脚=GPIO_PC(0),1	==>
背光名字="backlight_gpio0"	==>
☐ gpio_backlight1	
☐ gpio_backlight2	

< [root] - [xburst2系列cpu] - [x2000系列 cpu 设置]

SPI 总线 ==>

☒ LCD 驱动	==>
☒ lcd layer mixer(图像叠加)	
☐ rotator(图像旋转)	

< [root] - [xburst2系列cpu] - [x2000系列 cpu 设置] - [LCD 驱动]

☐ pan display 时等待帧同步完成(帧缓冲数为一帧时推荐开启)

☐ 强制使用默认的层级配置 (fb0为最底层,fb3为最顶层,应用配置无效)

frame desc 地址(用于保持开机logo,0表示不强行指定)=0 ==>

☒ 使能fb0	
fb0帧缓冲数量=2	==>
设备节点名字="fb0"	==>
☐ 使能fb1	

< [root] - [驱动] - [utils drivers(实用驱动列表)]

☒ gpio pin driver(结合gpio与有效电平的操作)

☐ 驱动内存继承接口

四 添加测试代码

vendor/vendor.c

```

1  #include <stdio.h>
2  #include "example/driver/isp_for_dualcamera_example.c"
3
4  void vendor_init(void *arg)
5  {
6      printf("vendor init...\n");
7      thread_test_isp();
8  }
```

编译烧录后将会在屏幕上看到sensor1预览到屏幕上的数据。同时在串口终端可以看到sensor0获取到的帧数：

```
U-Boot SPL 2013.07-00053-g1fc1a23-dirty (Jan 12 2022 - 18:56:14)
ERROR EPC 8ffac820
Current Version: V2 硬件环境
CPA_CPAPCR:0310086d
CPM_CPMPCR:07c0484d
CPM_CPEPCR:0310186d 配置sensor相关
CPM_CPCCR:9a094410
DDR: W63AH6NKB_BI type is : LPDDR3
DDRP_INNOPHY_CALIB_DELAY_AL:00000051
DDRP_INNOPHY_CALIB_DELAY_AH:00000051
----ddr_read1(DDRP_INNOPHY_CALIB_DONE): 00000003
275, VID=0x0000009b, PID=0x00000012
[0.000000] xburst2 rtos @ Dec 9 2025 16:11:44, epc: 8ffac820
[0.000025] spl argument boot is args is NULL
[0.000200] gpio: VDDIO_CIM(PA00~PA17) = 1.8V
[0.000352] gpio: VDDIO_SD(PE00~PE05) = 3.3V
[0.000590] Supported Nand Flash, AT025D1GA(9b:12)
[0.038887] vendor init...
[0.040538] mclk already enabled rate=23809523, not change to 24000000
[0.062614] channel = mscaler1-ch0
[0.062756] sensor_name = sensor1_gc2053
[0.062906] width = 720
[0.063019] height = 1136
[0.063136] fps channel = mscaler0-ch0
[0.063319] sensor_name = sensor0_gc2053
[0.063469] width = 720
[0.063582] height = 1136
[0.063699] fps = 30
[0.063809] data_fmt = NV12
[0.063925] line_length = 720
[0.064039] frame_size = 1226880
[0.064165] = 30
[0.064249] data_fmt = NV12
[0.064366] line_length = 720
[0.064479] frame_size = 1226880
[0.064606] frame_align_size= 1228800
frame_align_size= 1228800
[0.085309] ## VIC WARN status = 0x00001022
[0.122401] frame count: 1
[0.589211] backlight level: 1
[1.122391] frame count: 2
[2.122394] frame count: 3
[3.122394] frame count: 4
[4.122395] frame count: 5
[5.122392] frame count: 6
[6.122392] frame count: 7
[7.122392] frame count: 8
[8.122392] frame count: 9
[9.122393] frame count: 10
[10.122393] frame count: 11
```

☒ gpio pin driver(结合gpio)
☐ 驱动内存继承接口

四 添加测试代码

```
1 #include <stdio.h>
2 #include "example/d
3
4 void vendor_init(vo
5 {
6     printf("vendor
7     thread_test_isp
8 }
```

编译烧录后将会在屏幕上看到s